Category Theory and Functional Programming
Algebras and Recursive Data Types

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Preliminaries

Textbook

S. Abramsky and N. Tzevelekos (2011). Introduction to Categories and Categorical Logic.

Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

Outline

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Introduction

The relation between recursive (inductive) data types and initial algebras is described by (Fong, Milewski and Spivak 2020, p. 98) as:

> *"The string of a recursive data type is a functor, and the knot is its initial algebra."*

Functors Associated with Recursive Data Types

Example (Natural numbers)

We define the recursive data type for natural numbers by

```
data Nat = Zero | Succ Nat
```

and we define the non-recursive associated functor by

```
data NatF a = ZeroF | SuccF a

instance Functor NatF where
  fmap :: (a -> b) -> NatF a -> NatF b
  fmap _ ZeroF      = ZeroF
  fmap f (SuccF n) = SuccF (f n)
```

Functors Associated with Recursive Data Types

Example (Lists)

We define the recursive data type for lists by

```
data List c = Nil | Cons c (List c)
```

and we define the non-recursive associated functor by

```
data ListF c a = NilF | ConsF c a

instance Functor (ListF c) where
  fmap :: (a -> b) -> ListF c a -> ListF c b
  fmap _ NilF          = NilF
  fmap f (ConsF x y) = ConsF x (f y)
```

Outline

F-Algebras

Definition

Let $F : \mathcal{C} \to \mathcal{C}$ be an endofunctor. An F-**algebra** (A, α) is (Fong, Milewski and Spivak 2020, p. 104)

(i) an object A in $\mathrm{Obj}(\mathcal{C})$ (the **carrier**),

(ii) an arrow $\alpha : F_0\, A \to A$ (the **structure map**).

Algebra Arrows

Definition

Let $F : \mathcal{C} \to \mathcal{C}$ be an endofunctor and let (A, α) and (B, β) be two F-algebras. An **algebra arrow** $f : (A, \alpha) \to (B, \beta)$ is an arrow $f : A \to B$ in $\mathcal{C}$ such that the following diagram commutes (Fong, Milewski and Spivak 2020, p. 104):

$$\begin{tikzcd} F_0\, A \arrow[r, "F_1\, f"] \arrow[d, "\alpha"'] & F_0\, B \arrow[d, "\beta"] \\ A \arrow[r, "f"'] & B \end{tikzcd}$$

$$\big(f \circ \alpha = \beta \circ (F_1\, f)\big)$$

Outline

References

S. Abramsky and N. Tzevelekos (2011). Introduction to Categories and Categorical Logic. In: New Structures for Physics. Ed. by Bob Coecke. Vol. 813. Lecture Notes in Physics. Springer, pp. 3–94. DOI: 10.1007/978-3-642-12821-9_1 (cit. on p. 2).

Brendan Fong, Bartosz Milewski and David I. Spivak (2020). Programming with Categories (DRAFT). Last update: October 6, 2020. URL: http://brendanfong.com/programmingcats.html (cit. on pp. 5, 9, 10).